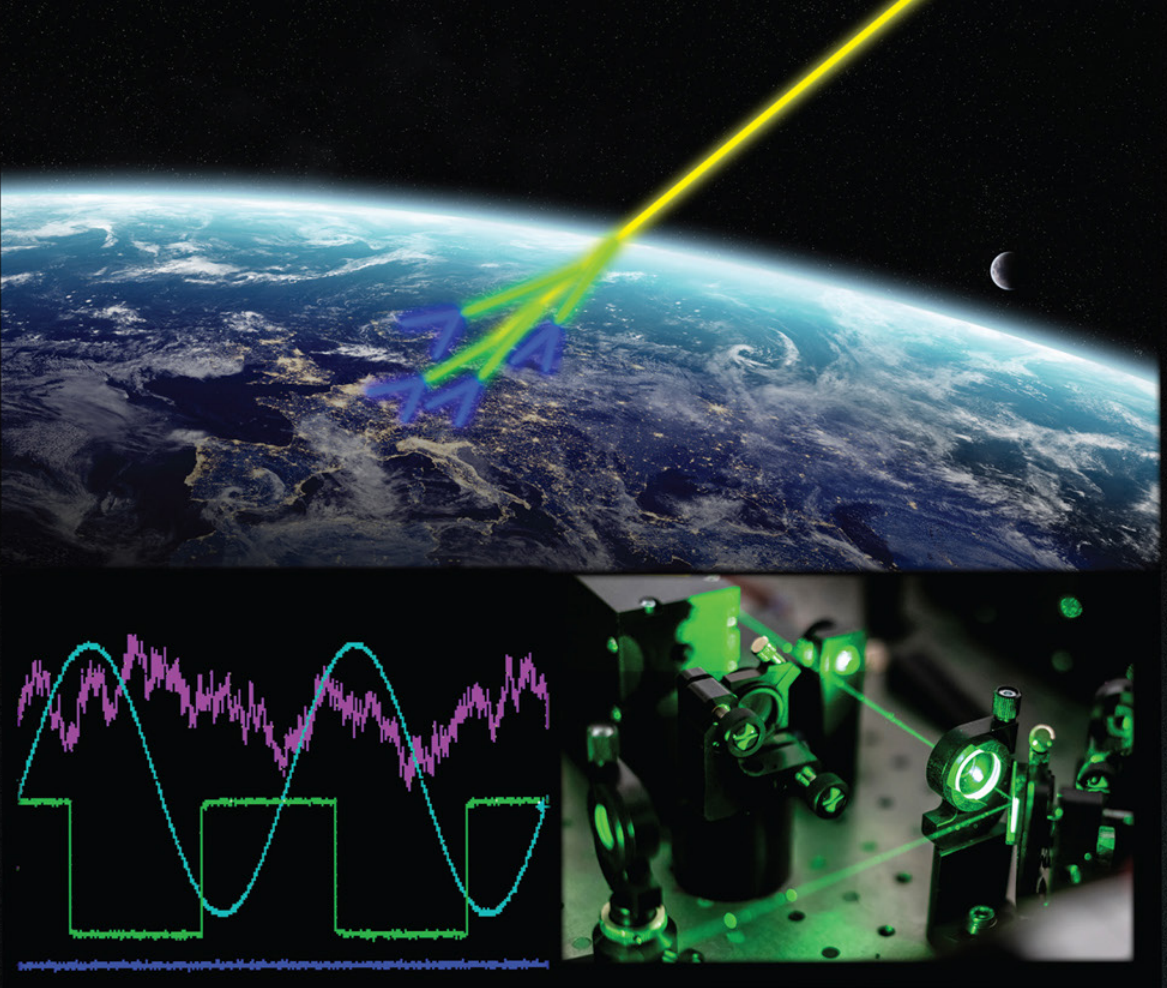




EXPERIMENTAL PHYSICS

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for the Laboratory

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4

Uncertainty and Statistics

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Although this may seem a paradox, all exact science is dominated by the idea of approximation. When someone claims to know the exact truth about anything, you are safe in inferring that this is an inexperienced person. Every careful measurement in science is always given with the probable error. Careful observers admit that they are likely wrong, and know about how much wrong they are likely to be.

– Adapted from Bertrand Russell, *The Scientific Outlook*

Suppose that you calculate the output of a circuit and predict an output voltage of 6 volts. You then build the circuit in the laboratory and your digital voltmeter reads 5.98 V. Does this measurement support your prediction or contradict it? The answer depends on your *uncertainty*.

4.1 Random vs. Systematic Errors

Uncertainty, also known as *error*,* is how scientists communicate their expectations about how closely their experimental result will match the results from past or future experiments, or match a theoretical prediction. Uncertainty is an essential part of every scientific prediction or measurement – it defines what we mean by “agreement,” making it possible for our experiments to confirm or disconfirm hypotheses.

Accuracy vs. Precision

There are two main ways that errors affect measurements: (1) by lowering the *precision* (statistical error) and (2) by lowering the *accuracy* (systematic error). Precision tells you how much your measurements look like one another; accuracy tells you how close they are to the correct value. No matter how careful you are in making your measurements, there will always be some unpredictable spread in your data due to random changes in variables beyond the experimenter’s control (e.g., air currents), from variations in how the experimenter takes readings from analog devices (e.g., meters with needle pointers rather than digital displays), and sometimes from variations in quantities that are intrinsically random (e.g., the path a particular photon follows). Suppose that you track the motion of a ball through the air. Even if the air is at rest on average, there are always air currents on small scales that can deflect the ball. Since the currents are just as likely to push the ball up as down, if you repeat the experiment many times, your data will look scattered or *imprecise*. By averaging the scattered data, you can hope to “average out” the air’s random influence and lower your uncertainty. The more data you have to average, the more you can reduce this kind of uncertainty (Figure 4.1).

The scatter due to imprecision does not reflect any *systematic* errors, the type of errors in procedure or equipment that make repeated measurements consistently high or low. Systematic errors cause a lack of *accuracy* in your measurement – no matter how many repeated measurements you take, systematic errors will cause them all to differ from the true value. In our ball example, while the air can push the ball either left or right, it will always push the ball backwards and not forwards: air resistance will slow the ball down. If we were to repeat the experiment multiple times this effect would always act in the same direction, so all the data would be shifted in a systematic way and the

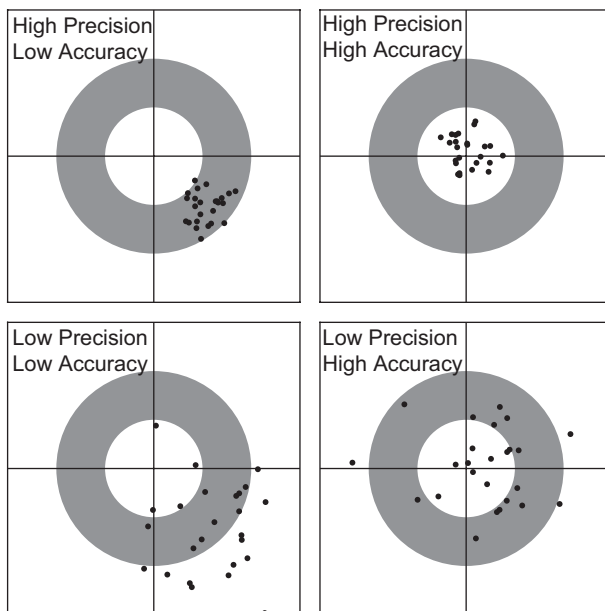


FIGURE 4.1 Results with different accuracy and precision. Note how improving the accuracy (left to right) produces a distribution that is better centered on the true value (center of the circle), while improving the precision (bottom to top) reduces the “scatter,” but without necessarily drawing closer to the correct value

* Although uncertainty is sometimes referred to as error, its presence is not necessarily an indication of some experimental mistake (as in “human error”). When we use “error” to mean uncertainty, we are referring to its alternative meaning of “wandering,” since the error is an estimate of how much our measured value will wander from our best estimate of its true value.

result will be *inaccurate*. Systematic errors can be much harder to quantify than statistical errors because you don't necessarily know that the data have been shifted, even if you repeat the experiment multiple times.

Where Do These Systematic Errors Come From?

They fall into a small number of general categories:

- *Offsets*: your measuring device reads an incorrect value that always differs from the true value by a fixed additive amount.
 - your voltmeter reads 0.2 V even when the probes are connected together
 - your meter stick is worn off at one end, reducing all your measurements
 - a pressure sensor came from the factory always reading 0.02 kPa below the actual pressure
 - your light sensor generates a small thermal current, even in the dark
- *Calibration errors*: your measurement device reads an incorrect value, with an error that depends on the input value.
 - your voltmeter reads 1% too high, so that it reads 1.01 V when you apply 1.00 V and 2.02 V when you apply 2.00 V
 - your meter stick expands slightly, reading 1 cm for every 1.01 cm of actual distance
 - a force sensor always reads $1.2 \times$ the actual force applied
 - a light sensor is designed to produce one digital count for every 3.6 photoelectrons, but instead produces one count for every 4.1 electrons

Note that calibration errors are often multiplicative, as in the above examples, but not always. For example, a current meter might read 1% too high for currents below 1 mA, changing gradually to 2% too high for currents above 5 mA.

- *Insertion errors*: your measurement affects the quantity to be measured in some way.
 - your thermometer heats up the sample as you measure its temperature
 - the addition of a force sensor changes the mass of your experiment
 - the electronics on your light detector give off infrared radiation that gets picked up by the light sensor
 - your current meter has a resistance that is significant compared with that of your circuit, so that when you put the meter into the circuit, the current flowing in the circuit is significantly reduced
- *Drift*: changes in experimental conditions not related to the phenomenon you are studying cause the experiment to change over time.
 - the lab room heats up during the day, changing your experiment's cooling rate
 - vibrations from vehicle and foot traffic disturb your experiment more during the day than at night
 - your radioactive sample decays during the experiment, inadvertently causing your detection rate to decline
 - impacts from cosmic rays degrade your detector over time, lowering its efficiency

Some of these systematic errors can be identified after the fact and their effects can be corrected or partially remediated, but others must be measured at the time of the experiment by frequently calibrating your measuring apparatus against a stable reference standard. This can be accomplished by checking your measurements against a separate measuring device that has been calibrated against an outside standard, or by applying your device to some well-understood physical system with standard properties;

for example, you might use the spectrum from a sodium emission lamp, which has a large number of well-measured spectral lines, to calibrate a spectrograph.

4.2 Methods of Determining Uncertainty

There are two basic methods for determining uncertainty: instrumental uncertainty and multiple trials.

Instrumental Uncertainty

Sometimes an experiment is so difficult or time-consuming that it can be performed only once, or your measuring apparatus makes such coarse measurements that repeated trials will return the same numerical value. In these cases, your uncertainty will be a direct result of the quality of your measuring apparatus, which you can either measure directly or obtain from some accepted source (such as other users or the manufacturer). If you need to make your own precision estimate, you can observe the variations of your measuring device while making some measurement that you expect to be stable; if you need to know your accuracy, you can apply your measuring device to a known standard. You can also get error values for instruments from the device manufacturer's specifications (which summarize their attempts to make the above observations). This method allows you to take both accuracy and precision into account when determining your uncertainty. We'll discuss this method in more detail in Section 4.9.

Multiple Trials

If the experiment can be repeated many times quickly, it is usually easiest to use the results of these multiple trials to estimate the precision. This method does not measure the accuracy; for example, it won't detect a calibration error. If you don't know anything about the accuracy of the instruments, you might proceed under the assumption that the uncertainty is dominated by the lack of precision, rather than the lack of accuracy. But in that case, if you get a statistically significant difference between your measurement and the expected value, that might be a true discrepancy (an important result!) or it might merely mean that your uncertainty is not primarily statistical, but systematic instead.*

4.3 Standard Error of the Mean and Probability Distributions

(Much of the following discussion is phrased in terms of the multiple trials method, but as we'll see it also applies to the instrumental uncertainty method.)

4.3.1 Sample vs Population and the Gaussian Distribution

When you go into the laboratory, you perform a finite number of measurements; this set of measurements is your "sample." You might be measuring a speed, a voltage, or some other physical property; this is your variable, x . However, the result contains some uncertainty – the readings are not always identical. Often, you'll report the mean $\mu_s = \bar{x}$ of your sample, i.e., the average of your multiple measurements. To characterize the uncertainty in this reported mean, you will use the "Standard Error of the Mean," or SEM. To understand this fully, you need a bit of statistics background and nomenclature.

Think of your sample as being drawn from an infinitely large pool of possible results; this pool is called the "population." The range and frequency of the values in the population are specified by a "probability distribution." The probability distribution is a function, $P(x)$, of your variable, and the integral of

* A related tool for estimating statistical uncertainty is *resampling*, in which you simulate multiple hypothetical samples (usually with the aid of a computer) from your one actual data set to gauge the effects of outliers and sample size.

$P(x)$ from x to $x + dx$ gives the probability that a value in that range will show up in a single measurement. Since every measurement returns some value, the probability is normalized to have integral 1 over all possible measurement values.

The vast majority of laboratory measurements that are not counting experiments (e.g., counting radioactive decays) follow a “normal” (or “Gaussian”) distribution. The normal distribution has a *mean*, or average value, that is also its *mode*, or most popular value, and the likelihood of getting a different value falls off rapidly on either side of the mean according to

$$P(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (4.1)$$

where μ is the population mean and σ is the “standard deviation,” which measures the breadth of the distribution (Figure 4.2).

The standard deviation describes the variation of values in the population, or equivalently it tells you how wide $P(x)$ is. It can be estimated from your measurements using*

$$\sigma = \sqrt{\frac{\sum_i (x_i - \mu_s)^2}{N - 1}} \quad (4.2)$$

where $\mu_s = \bar{x}$ is the mean of your sample, a sample which is assumed to contain N independent measurements drawn from the population. This is only an estimate for σ because, due to your finite number of measurements, you don’t know the true mean μ of $P(x)$, i.e., the mean of the population. Instead, you only

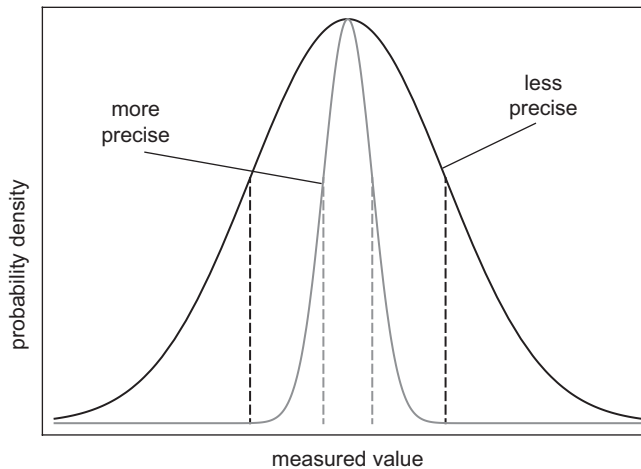


FIGURE 4.2 Two normal distributions with different widths. The darker line plots a distribution with a large standard deviation, while the lighter line shows a smaller standard deviation. The standard deviation is indicated in both cases by a pair of dashed lines enclosing 67% of the probability.

* This is simply a way of measuring the typical deviation from the average. Starting from the inside of the expression, we take the difference between each data point and the average. We square this so that both positive and negative differences from the average count in the same way. Then (with the summation and the division by $N - 1$), we essentially take the average. (See the next footnote for an explanation of why we use $N - 1$ rather than N .) Finally, we take the square root so that the result has the same units as x .

know the mean of your sample μ_s . However, if you have at least 15 measurements, it's usually a good enough estimate. (See Section 4.5 if you have fewer than 15 measurements.*)

4.3.2 Standard Deviation vs. Standard Error of the Mean

If you take 5000 measurements instead of 50 measurements, your estimate for σ will be about the same; you're just getting a more accurate idea of the variations in the population. However, your sample mean μ_s is a much better estimate of the true mean μ , so the uncertainty in your reported mean is much lower.

The uncertainty in μ_s is the SEM, which you can estimate using:[†]

$$\text{SEM} = \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{N}} \quad (4.3)$$

The standard error of the mean now properly decreases with a larger sample size.

4.3.3 Other Distributions

The normal distribution does a good job of describing data sets in which the total error is the sum of many smaller contributing errors,[‡] which is why it is so common and useful in the physical sciences. Other common distributions include the *Poisson distribution*, which describes the uncertainty in a discrete count of events, and the *binomial distribution*, which describes the uncertainty in a set of independent binary observations, such as a series of coin flips. Poisson statistics are discussed in detail in Section 13.4.

4.3.4 Median and Mode

Example: Characterizing Voltage Data

We test a battery's voltage by hooking up a voltmeter to its terminals. After repeating this procedure 20 times, the results are as follows: 1.40, 1.41, 1.43, 1.43, 1.44, 1.44, 1.44, 1.45, 1.45, 1.45, 1.46, 1.46, 1.47, 1.47, 1.48, 1.48, 1.50, 1.51. What are the *mean*, *median*, and *mode* of these data? What is the standard deviation and what is the standard error on the mean?

Solution: The mean of an unweighted data set can be calculated from

$$\mu_s = \frac{\sum x_i}{N} \quad (4.4)$$

which for this data set gives us $1.4535 \approx 1.45$ V. This is the mean that is returned by Excel's AVERAGE function or *Mathematica*'s Mean[] function.

The median is the "middle" value of the data set, the value that is larger than half of the measurements and smaller than the other half. The median is sometimes preferred to the mean

* If $N - 1$ were replaced with N , the quantity under the square root would be the average squared deviation of the measurements from the mean; by using $N - 1$, we increase the estimate of σ for small samples. Since the members of your sample are likely to be closer to their own sample mean than to the true population mean (i.e., $\sum (x_i - \mu_s) \leq \sum (x_i - \mu)$ where μ is the true population mean, not the mean of your particular sample), the factor of $N/(N - 1)$ helps to correct what would otherwise be a low-biased estimator. Another way to think of this: because you had to use your sample to determine the mean, you have only $N - 1$ independent measurements of how the sample is scattered around the mean.

[†] Please see the entry for this section on ExpPhys.com for the derivation of this formula.

[‡] This observation is known as the *central limit theorem*.

because it is less easily changed by *outliers*, data that lie at a great distance* from the mean. Since this data set has 20 values, the middle falls between the 10th and 11th values, so the standard procedure is to take the mean of values 10 and 11. Both of those values are 1.45 for this data set, so the median (like the mean) is 1.45 V.

The mode is the value that appears most frequently in the data. We can visualize the data using a histogram (Figure 4.3) and identify the most common value as 1.44, with four occurrences. No other value† appears more than three times, so 1.44 is our mode. For probability distributions that are very different from the normal distribution, the mode might do a better job than the mean of telling you what to expect in future experiments.

Finally, we should estimate the standard deviation and the standard error of the mean. We get the standard deviation from summing the squared differences from the sample mean and dividing by $N-1$. The sample mean is 1.454, so:

$$\sigma \approx \frac{(1.40 - 1.454)^2 + (1.41 - 1.454)^2 + 2 * (1.43 - 1.454)^2 + \dots}{(20 - 1)} = 0.03$$

This standard deviation can be computed with Excel's STDEV.S function or *Mathematica*'s StandardDeviation function.

The standard error of the mean, on the other hand, is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.027}{4.47} = 0.006$$

which is much smaller than the standard deviation.

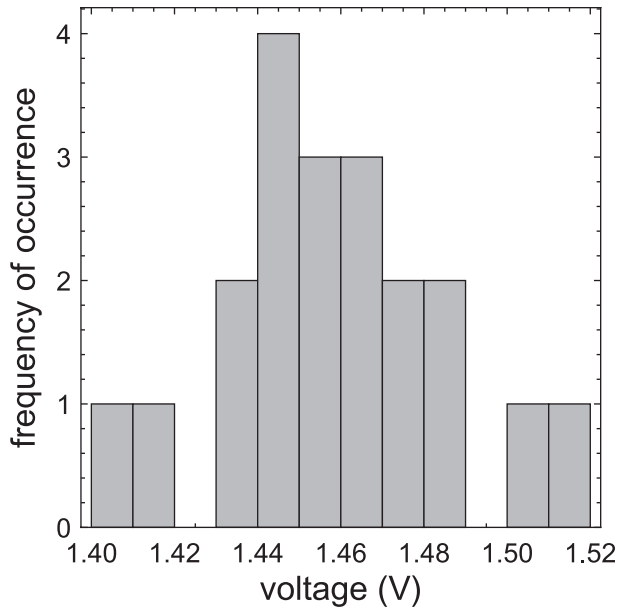


FIGURE 4.3 A histogram of the voltages from the example. The height of each bar represents the number of times that the corresponding value appeared in the data set. The mode can be identified by finding the value with the most appearances, which for these data is 1.44 V.

4.4 Confidence Intervals

You should report results as the mean of the measurements and the width of a specified *confidence interval*. This is defined as a range, Δ , of your measurement variable that contains the mean, μ_s , and has the property that

* Determining what constitutes a “great distance” and how to properly account for outliers without allowing our biases to overrule the actual data are difficult issues to be discussed in Section 4.11.

† This single most common value makes our data *unimodal*; if there were more than one widely separated value that dominated over its neighbors, the data would be *multimodal*.

$$\int_{\mu_S - \Delta}^{\mu_S + \Delta} P(x) dx = C \quad (4.5)$$

where C is a probability threshold that you must choose. Common values for C include 68%, 95%, and even as high as 99.9999%, which is typical for particle physics, where a premium is placed on avoiding false positive results. *It is critical that you specify the value of C the first time you quote a result with uncertainty. Alternatively, you can quote the value of C as a percentage in parentheses, as in the examples below.* You should use 95% in most cases.

For the specific case of the normal distribution, if the true population mean is μ and you make 15 or more independent observations and average them to get $\bar{x} = \mu_S$, then

- 68% of the time, your measured $\bar{x}$ will fall within the range $\mu \pm \sigma_{\bar{x}}$
- 95% of the time, your measured $\bar{x}$ will fall within the range $\mu \pm 2\sigma_{\bar{x}}$
- 99.7% of the time, your measured $\bar{x}$ will fall within the range $\mu \pm 3\sigma_{\bar{x}}$,

where again $\sigma_{\bar{x}}$ is the standard error of the mean. So, if you're using a 95% confidence interval, then you will report your result as $\bar{x} \pm 2\sigma_{\bar{x}}$ (95%).

You must be careful to use the term “confidence interval” (CI) properly. For example, say you measure the gravitational acceleration to be $(9.83 \pm 0.02) \text{ m/s}^2$, where the $\pm 0.02 \text{ m/s}^2$ is the 95% CI. The expected result of 9.81 m/s^2 is right at the edge of the 95% CI. It would be correct to say, “The accepted value is within the 95% confidence range of my experiment, indicating a reasonable level of agreement.” However, because the expected value is right at the edge of the CI, it would also be correct to say, “There is only a 5% chance that the discrepancy between my measurement and the expected value was caused by fluctuations and inaccuracies that are included in my uncertainty analysis.” It would be **wrong** to say, “My results show with 95% confidence that the expected value is correct.” After all, if you had chosen instead to quote a 99.7% CI and the expected value was just at the edge of this CI – i.e., the discrepancy is greater than above – you would certainly not say, “My results show with 99.7% confidence that the expected value is correct.” In this case, there is 99.7% chance that the discrepancy was caused by a systematic error in your experiment that is not accounted for in your uncertainty analysis, or that the accepted value is wrong.

4.5 Student's t-Distribution

For fewer than 15 measurements, your confidence intervals should be broader than those quoted here, due to additional uncertainty caused by your small sample. If we assume that your measurements represent an independent sample randomly drawn from a normally distributed variable with a well-defined mean and standard deviation, then your sample means should follow “Student's t-distribution.”* This distribution has one parameter, ν , which specifies the number of “degrees of freedom”:

$$\nu = (\text{\# of independent measurements}) - (\text{\# of fitted or calculated properties}) \quad (4.6)$$

The Student's t-distribution is symmetrical, like the normal distribution, but has more probability farther out from the mean, and thus a higher likelihood of producing samples with means that are very far from the underlying population mean (Figure 4.4). This extra width is more pronounced for lower values of ν . The mathematical form for the t -distribution is complex, but it can be accessed directly in *Mathematica* using the `StudentTDistribution[ $\nu$ ]` function. The range, measured in standard deviations, that must be included to produce a given % CI can be calculated in Excel using the function `T.INV.2T(1–C.I.,  $\nu$ )`. Alternatively, you can use Table 4.1.

* This was developed by William Sealy Gossett, Head Brewer of Guinness and an important figure in modern statistics. He published under the pseudonym “Student.”

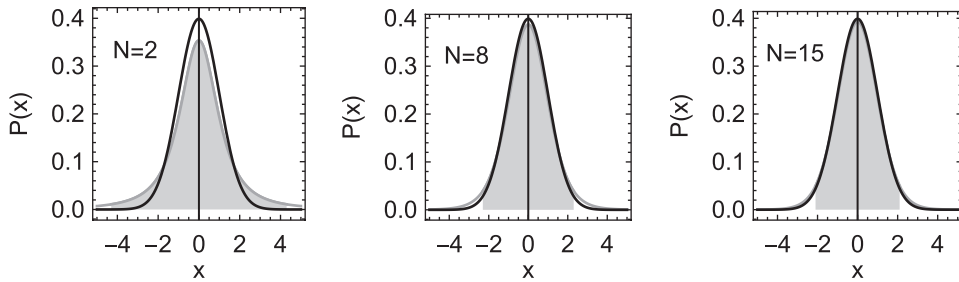


FIGURE 4.4 Normal (black line) and Student's t (gray line) distributions for 2, 8, and 15 degrees of freedom. The x coordinate is measured in standard deviations (σ). The shaded areas show the range enclosing 95% of the probability, which approaches 2σ as the sample gets larger.

TABLE 4.1

Factors for converting standard error to confidence interval using Student's t -distribution. Find the number of degrees of freedom in the first column and multiply your standard error by the factor under the desired confidence interval

η	Probability contained		
	67%	95%	99%
1	1.75	12.71	63.66
2	1.28	4.30	9.92
3	1.16	3.18	5.84
4	1.11	2.78	4.60
5	1.08	2.57	4.03
6	1.06	2.45	3.71
7	1.05	2.36	3.50
8	1.04	2.31	3.36
9	1.03	2.26	3.25
10	1.02	2.23	3.17
11	1.02	2.20	3.11
12	1.02	2.18	3.05
13	1.01	2.16	3.01
14	1.01	2.14	2.98
15	1.01	2.13	2.95

Example: Using Student's T-Distribution

Suppose that you are running an experiment to measure the resistance of a sample, collecting seven different values of current as you apply seven different voltages. You fit a line to the data and find that the best-fit slope corresponds to $50 \pm 5 \, \Omega$, where $5 \, \Omega$ is the standard error (the equivalent to the standard error of the mean, but the result of a fit) reported by your fitting software. How should you take your number of data points and fitted parameters into account if you want to report the 99% confidence interval?

Assuming that your seven data points are independent, the fit parameters should be distributed according to Student's t -distribution for the appropriate number of degrees of freedom. Here, we have seven data points minus two fitted parameters (slope and intercept of a line) = 5 degrees of freedom for our distribution. We wanted to report a 99% confidence interval, so we need to know how wide that is, in standard errors, for a Student's t -distribution with 5 degrees of freedom. In Excel, we use $T.INV.2T(1 - 99\%, 4) = 4.03$. Then, $4.03\sigma_{\bar{x}} = 4.03 \times 5 \, \Omega = 20.15 \, \Omega$. So for this data

set, we can report that the resistance is $50 \pm 20 \, \Omega$ at 99% confidence, or simply $50 \pm 20 \, \Omega$ (99%). Why 20 and not 20.15? Because we don't want to overstate our confidence in the error value by reporting too many significant figures.

4.6 Significant Figures

We can use the confidence interval to find the correct number of significant figures for expressing our results without giving a false impression of our precision or accuracy. First, determine the correct number of significant figures for your uncertainty itself. Since we've seen that the standard error of the mean decreases as $\sqrt{n}$, where n is the number of measurements, then a safe assumption would be that

$$\frac{\text{uncertainty}_x}{x} \geq \frac{1}{\sqrt{n}} \quad (4.7)$$

So if we have tens or hundreds of measurements, we can justify rounding the uncertainty to the first significant figure, since that will correctly convey that we have $\approx 10\%$ error *on our uncertainty value*. The one exception would be if the uncertainty value itself begins with a 1, since rounding 14 to 10 (or 1400 to 1000) introduces an error larger than our 10% error estimate. So in the case where the first digit of our uncertainty is a 1, we can justify keeping one additional digit. If we are running a very large experiment that takes 10,000 data points or more, we can consider keeping two or more significant figures in our uncertainty, always rounding to the decimal place corresponding to $1/\sqrt{n}$.

Now that we have a properly rounded uncertainty value, we can let it guide us in how to round our actual measured value. There is no point in reporting a temperature of 22.56°C from a thermometer with an uncertainty of $\pm 2^\circ\text{C}$, and it might mislead future experimenters. To keep our value consistent with the uncertainty, we should round the value to the same decimal place as our uncertainty. In the thermometer example, the error is rounded to the units place, so the measured temperature should also be rounded to the nearest $^\circ\text{C}$, for a reported value of $23 \pm 2^\circ\text{C}$.*

4.7 Quantitative Comparisons, or How Not to Be Misled by Error Bars

Once we have determined the correct confidence range, we are ready to test for agreement between our result and other results or theory. For example, we might be testing whether a resistor has the correct value to match its label by applying different voltages and measuring the currents, then determining the slope of the line. If we find $550 \pm 30 \, \Omega$ (95%), but the theoretical value[†] is $500 \pm 25 \, \Omega$ (95%), is our value in agreement with the specified value?

In this example, the error bars from the measured and theoretical values overlap (see Figure 4.5), but that does **not** mean that our value agrees with the theoretical value at the 95% level. The correct way to check for agreement is to determine whether

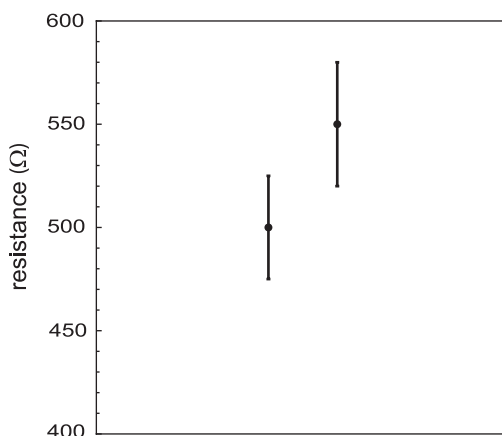


FIGURE 4.5 A graphical representation of the theoretical and measured resistances from the example, with uncertainties shown as error bars. Note that the error bars overlap in this example, with the higher end of the error range from the specified value exceeding the lower end of the error range for the measured value.

* We've chosen to report the central value, then the one-sided confidence interval, then the units. Although you might see other formats, this one has the advantage of communicating the value, error, and units without repetition or extra brackets.

[†] Theoretical values can have uncertainties, for example, because one of the parameters used in the theory is uncertain.

$$(x_{\text{measured}} - x_{\text{theoretical}})^2 \leq \text{uncertainty}_{\text{measured}}^2 + \text{uncertainty}_{\text{theoretical}}^2 \quad (4.8)$$

Using the example values, we find that the *discrepancy*, the difference between the measured and theoretical values, is $50 \, \Omega$, while the joint uncertainty on the discrepancy is $\sqrt{25^2 + 30^2} \, \Omega \approx 40 \, \Omega$, so despite those overlapping error bars, these two values do not agree at the 95% confidence level, meaning that we expect such discrepant values to indicate true disagreement at least 95% of the time.[†]

4.8 Propagating Errors

So far, we have found the errors in measured quantities, either by testing them ourselves or by referring to manufacturer's specifications (or other customary rules for estimating instrumental errors). But most experiments are not constructed so that your most interesting or useful quantity is directly measured in the experiment: you might measure speed when what you really want is kinetic energy, or measure the voltage across a resistor as a way of measuring the current. We need a way to get the error on that calculated quantity, based on the measured errors of the contributing quantities. This process is called *error propagation*.

Suppose that you have measured the speed of a $0.15 \, \text{kg}$ cart to be $0.30 \pm 0.05 \, \text{m/s}$ (95%). Its kinetic energy, $K = \frac{1}{2}mv^2$, is plotted in Figure 4.6. The central value is

$\frac{1}{2}(0.15 \, \text{kg})(0.30 \, \text{m/s})^2 = 0.0068 \, \text{J}$. There are

a few valid ways to estimate how much your $0.05 \, \text{m/s}$ error will affect your determination of the kinetic energy.

Direct Substitution

If you want to know how much the function $\frac{1}{2}mv^2$ changes when you change v by $0.05 \, \text{m/s}$, you could simply plug in your new value and see how much the function changes, that is

$$\begin{aligned} \Delta K(v) &= K(v + \Delta v) - K(v) \\ &= K(0.30 \, \text{m/s} + 0.05 \, \text{m/s}) - K(0.30 \, \text{m/s}) = 0.0024 \, \text{J} \end{aligned} \quad (4.9)$$

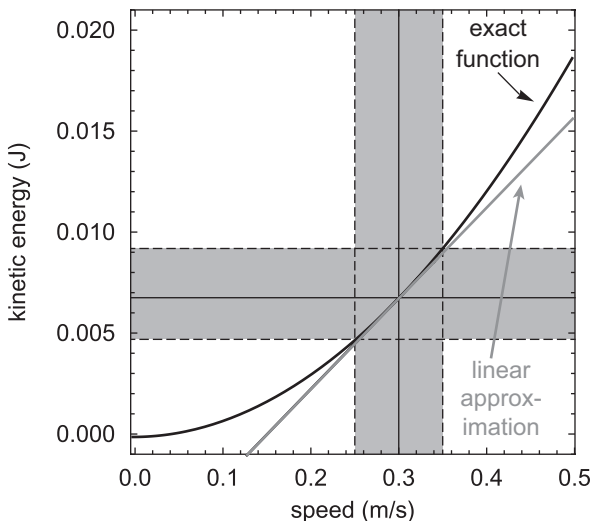


FIGURE 4.6 Kinetic energy of the cart as a function of speed. The gray areas span the error on the x-axis ($0.05 \, \text{m/s}$) and the error on the y-axis, as determined by evaluating the kinetic energy at the highest and lowest speed values given by the error bar. The straight line shows what we would obtain by using the derivative of the energy at the measured value ($0.30 \, \text{m/s}$) and extrapolating to those same error limits.

* Note that we can combine two uncertainties by adding in quadrature whether they represent $1\sigma_{\bar{x}}$ or $2\sigma_{\bar{x}}$ or any other confidence interval, so long as they both represent the same % confidence.

[†] The other 5% of the time, the apparent disagreement is accidental.

where ΔK is the uncertainty in the kinetic energy, with the same % confidence as for the uncertainty in velocity Δv . This method has the advantage that it correctly models the higher-order behavior of the function, which can be a substantial improvement for calculated quantities that are highly non-linear. It does not address a related problem: although adding 0.05 m/s to our measurement gives one estimated error, subtracting it from our measurement will give a slightly different error (0.0021 J, in this case) with no reason to prefer one result over the other. For our purposes, however, this is not an important distinction. The error is seldom known to better than 10% in most research, so regardless of whether we add or subtract the error, we would report 0.007 ± 0.002 J (95%), rounding the error to one significant figure and then rounding the central value to match that decimal place. If you find that the error estimate from adding differs from the one you get by subtracting, you may average the two, or take the larger of the two (to be conservative) or you may choose to report both, as in $0.0068^{+0.0024}_{-0.0021}$ J (95%).

Linear Approximation

Rather than using the exact functional form of the calculated parameter, we might choose to approximate the change using its derivative, measured at the location of the central value. In our example:

$$\Delta K = \Delta v \left| \frac{\partial K}{\partial v} \right|_{v=\mu} = (0.05 \text{ m/s})(0.15 \text{ kg})(0.3 \text{ m/s}) = 0.0023 \text{ J}$$

This is indistinguishable from the direct substitution estimates, since only the 0.002 J would be reported, so for errors on functions of low order or for small fractional errors, this is a perfectly adequate way to propagate errors.

Multiple Error Contributions

In the kinetic energy experiment, we had two measured quantities that contributed to the energy: mass and speed. So far, we've been neglecting the error in the mass. This is appropriate as long as the fractional error on the mass is at least three times smaller than the fractional error on the speed, that is

$$\frac{\sigma_m}{m} \bigg/ \frac{\sigma_v}{v} \leq 1/3 \quad (4.10)$$

An error three times smaller than the next largest error cannot affect the total error at the 10% level, which is the usual level of significance for lab exercises and most research results.

Suppose that the mass error were larger, comparable with the fractional error in the speed, for example, $m = 0.15 \pm 0.04$ kg (95%). How can we include this error in the error propagation?

Addition in Quadrature

For a function $y(x, z)$, if we assume that variations in z are uncorrelated with variations in x , then:*

$\sigma_{y,\text{total}} = \sqrt{\sigma_{yx}^2 + \sigma_{yz}^2}$, where σ_{yx} is the uncertainty in y due only to variations in x , and σ_{yz} is the uncertainty in y due only to variations in z .

Addition in quadrature

* Please see the derivation of this formula under this section at ExpPhys.com

This way of adding uncorrelated error sources (squaring, adding, then taking the square root) is called “addition in quadrature.”

Returning to the kinetic energy example, we have

$$\sigma_{Kv} = K(m, v + \sigma_v) - K(m, v) = 0.0024 \text{ J}$$

$$\sigma_{Km} = K(m + \sigma_m, v) - K(m, v) = 0.0018 \text{ J}$$

$$\sigma_{K, \text{total}} = \sqrt{\sigma_{Kv}^2 + \sigma_{Km}^2} = \sqrt{0.0024^2 + 0.0018^2} \text{ J} = 0.003 \text{ J}$$

The resulting error bar of 0.003 J takes into account both the mass and speed errors, and is larger than either one alone.

In the linear approximation method, this would be

$$\sigma_{K, \text{total}} = \sqrt{\left(\sigma_v \frac{\partial K}{\partial v}\right)^2 + \left(\sigma_m \frac{\partial K}{\partial m}\right)^2}$$

More generally, for a function $y(x, z)$,

$$\sigma_{y, \text{total}} \cong \sqrt{\left(\sigma_x \frac{\partial y}{\partial x}\right)^2 + \left(\sigma_z \frac{\partial y}{\partial z}\right)^2} \quad (4.11)$$

You can show in problem 10 that if $y = Ax^n z^m$, then

$$\frac{\sigma_{y, \text{total}}}{y} \cong \sqrt{\left(n \frac{\sigma_x}{x}\right)^2 + \left(m \frac{\sigma_z}{z}\right)^2} \quad (4.12)$$

This shows that the importance of each measured error depends strongly on the power to which that quantity is raised. For our kinetic energy example, we get

$$\left(\frac{\sigma_{K, \text{total}}}{K_{\text{total}}}\right) = \sqrt{\left(\frac{\sigma_m}{m}\right)^2 + \left(2 \frac{\sigma_v}{v}\right)^2} = \sqrt{\left(\frac{0.04}{0.15}\right)^2 + \left(2 \frac{0.05}{0.3}\right)^2} = 0.43$$

Using this fractional error estimate gives $\sigma_{\text{tot}} \approx 0.43 \times 0.0068 \text{ J} = 0.0029 \text{ J} \approx 0.003 \text{ J}$, which is essentially identical to the 0.003 J estimate obtained from the direct substitution method.

4.9 More of the Instrumental Uncertainty Method, Including “Absolute Tolerance”

All the above discussion applies both to the multiple trials method of determining uncertainties and to the instrumental uncertainty method.* In the multiple trials method, the “population” from which your sample is drawn is the set of possible results in the absence of any systematic errors; this method gives you no information about systematic errors. So, when you find a significant discrepancy between theory and experiment (e.g., the theoretical value is outside the 95% confidence range), you can conclude that, very likely, there is either an error in the theory or a systematic error in the experiment. This should cause you to re-examine both the theory and the experiment for sources of the discrepancy.

* The one exception is the Student’s t-distribution, which is only relevant for the multiple trials method and for quantities derived from fitting to a plot.

In contrast, the uncertainties for instruments normally do include allowance for calibration and offset errors,* so these types of errors are included in the population, which is represented by $P(x)$. The population and $P(x)$ represent an infinite set of measurements, each one taken with a different set of instruments (so that the calibration and offset errors are different). This population includes effects of random errors as well as calibration and offset errors. So, using this method, when you find a significant discrepancy between theory and experiment, you can be reasonably confident that it is not due to calibration or offset errors. It could still be due to other types of systematic errors in the experiment, or to problems with the theory.

From the information provided by the instrument manufacturer, we can estimate the standard deviation of the population, i.e., we can estimate σ . If you make only one measurement (which is typical for the instrumental uncertainty method), then the number of trials is $N=1$, so for purposes of the preceding discussions, the “standard error of the mean” is $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{N}} = \frac{\sigma}{\sqrt{1}} = \sigma$. For example, if you use a balance

with an uncertainty of $\sigma=0.05$ g and you measure a mass of 50.00 g, then there is a 68% chance (including effects of typical random errors associated with the instrument, calibration errors, and zero offset errors) that the true mass is in the range 49.95 g to 50.05 g.

Typically, manufacturers quote an “absolute tolerance” for the accuracy. For example, a digital multimeter (DMM) may quote the accuracy of voltage readings as a “% of reading + counts,” such as “0.5% + 2.” The “counts” refer to the last digit displayed. For example, if the meter displays 1.56 V, the accuracy would be $0.5\% \times 4.56 \text{ V} + 2 \times 0.01 \text{ V} = 0.0228 \text{ V} + 0.02 \text{ V} = 0.0448 \text{ V}$. This means that the manufacturer of the DMM guarantees that the reading is correct to within 0.0448 V, i.e., that the actual voltage is somewhere in the range $1.56 \pm 0.0448 \text{ V}$.

How do we interpret this in terms of σ ? The manufacturers usually achieve these accuracies by individual testing of each unit, with the failed ones being sent back for parts replacement and re-testing. This means that, if we make the same measurement with many different copies of the same DMM, we will get a non-normal distribution. The details of the distribution depend on the testing protocol of the manufacturer, which is proprietary. So, all we really know is that, to essentially 100% confidence, the true voltage is in the range $1.56 \pm 0.0448 \text{ V}$.

We follow the advice of the National Institute of Standards and Technology (NIST) to estimate σ for such cases:

$$\sigma = \frac{\text{Absolute tolerance}}{\sqrt{3}}, \quad (4.15)$$

where the “absolute tolerance” is half the guaranteed range, i.e., the absolute tolerance is 0.0448 V in our example. So, in our example $\sigma = \frac{0.0448 \text{ V}}{\sqrt{3}} = 0.026 \text{ V}$, where we’ve rounded to two significant digits.

You should try hard to find the manufacturer’s specification (“spec”) for the accuracy, which can usually be found in the manual on the “specifications” page. Often, you can find the manual on the web.

However, if you’re unable to find the spec, as a second choice you can use the following rules of thumb.

For analog instruments (i.e., instruments without digital displays such as meter sticks, calipers, micrometers, volumetric flasks, needle and dial scales on pressure gauges), the total uncertainty (including effects of imprecision and inaccuracy), σ , can be estimated as half the smallest division. For example, if a meter stick is marked with 1 mm as the smallest division, then $\sigma=0.5$ mm. Obviously, you can (and should) read the meter stick to a greater precision (you should be able to read it to within 0.1 mm); the 0.5 mm figure includes inaccuracies in the calibration of the instrument, effects of thermal expansion, etc., as well as the imprecision in your ability to read it.

* Be aware that uncertainties quoted by an instrument manufacturer are usually only guaranteed for one year after purchase. After this, the zero offset and calibration errors may increase. So, when possible, you should check the zero offset and calibration. If that is not possible, as long as the instrument is 15 years old or less, you can probably get away with using the quoted uncertainty.

For digital instruments, the accuracy of the instrument is often much lower than you might expect from the number of digits being displayed. However, it is usually safe to use σ =(five times the value of the least significant digit place displayed). For example, if a voltmeter displays a reading of 36.93 V, you would use $\sigma=5 \times 0.01 \text{ V}=0.05 \text{ V}$.

In some cases (e.g., if you are forced to use an instrument in a less than optimal way, or if you are using a non-standard instrument), you will need to estimate the measurement uncertainty yourself, using a “seat of the pants” method. The best way of doing this is to estimate the 95% confidence interval (corresponding to $\pm 2 \sigma$): what is the highest number that would be reasonable to record for the measurement? What is the lowest? This range corresponds approximately to the 95% confidence interval. The span of this interval, from highest to lowest, is equal to 4σ (i.e., $\pm 2 \sigma$).

As described in Section 4.8, you can combine the errors from multiple instruments by adding them in quadrature, as long as you can assume the errors are not correlated with each other.

An important exception to this rule for combining errors occurs when two measurements have been made with the same instrument and they have given very similar results. For example, if you want to measure the pressure drop along a length of tube, you might choose to use the same barometer to measure both pressures, one after the other. The change might be relatively small, say from 101.5 kPa to 100.8 kPa. In that case, you might be justified in assuming that the systematic errors of the device are approximately the same for both measurements, in which case the difference $\Delta P=P_2 - P_1$ does not depend on those systematic errors, which cancel one another out. The only errors that would affect ΔP would be random errors, which will be less than the combined effects of random and systematic errors. If you encounter a situation like this in your lab, where you are using the same meter to make multiple similar measurements, you should consider trying to characterize the precision separately from the total instrument error.

4.10 Parameter Fitting

In many experiments, you will not be exploring in complete ignorance, but will have some set of expectations that you want to adduce evidence for or against. In those cases, you will test your data against a *model*, a set of assumptions based on previous observations that allow you to predict future observations. A model might include a functional form for how one variable depends on another or go farther and predict one or more constants of the fit based on known constants of nature, while leaving some parameters unknown. In order to determine whether your model has done a good job of explaining your data, it is often necessary to determine the *best fit* values, the numerical values that bring the predicted functional form into best agreement with your data set, and then compare those with other experimenters’ best fit values or to values calculated using constants of nature.

The most common way to find these best fit values is by using *least squares* fitting. Your model should make a prediction for each observed case, whether it’s a large number of repeated, identical measurements (like multiple observations of the same star’s brightness) or a functional relationship (like the relation between current and voltage for a resistor). Subtracting your observed values from those predictions gives a set of differences that we call *residuals*. Least squares fitting changes your model parameters to minimize the sum of the squared residuals, or

$$r_{fit}^2 = \sum_{i=1}^n (x_i - m_i)^2 \quad (4.16)$$

where m_i is the model’s prediction for observation i . This expression works well for data sets where each point is of equal quality, or where the errors for each data point are not well known. In these cases, a model’s explanatory power can be expressed as an R^2 or *residual sum of squares* value. Taking the original data set, we can calculate the sum of squares using the mean of the data set (a zero-parameter fit, as it were):

$$r_{\text{mean}}^2 = \sum_{i=1}^n (x_i - \mu_s)^2$$

$$R^2 = 1 - r_{\text{fit}}^2 / r_{\text{mean}}^2 \quad (4.17)$$

The residual sum of squares (“R-squared”) expresses how much of the variance of the original data set has been explained by the model. Values range from zero to 1, with values above 0.9 usually indicating a good model fit. However, R^2 tends to overestimate how much of the variance is explained by the model because it is built from two biased estimators for the variance. Using better estimators gives the “adjusted R-squared,” given by

$$R_{\text{adj}}^2 = 1 - \left(1 - R^2\right) \frac{n-1}{\nu-1} \quad (4.18)$$

where ν is the number of degrees of freedom (Equation 4.6).

In *Mathematica*, least squares fitting is implemented by the `LinearModelFit*` command. This will return a `FittedModel` object that can be plotted directly or queried for parameters like R^2 . In Microsoft Excel, least squares fitting can be obtained by inserting a scatter plot of your data, right-clicking on the data points, and selecting Add Trendline from the context menu. Option checkboxes in the Trendline dialog box will allow you to display R^2 and the fit parameters.

Example: Polynomial data: We can use linear least squares fitting to analyze the simulated data set plotted in Figure 4.7. If we were to approach this data set with no preconceived idea of the correct functional fit to use, we might still notice that there is a pronounced trend for values at the right of the plot to be higher than values at the left of the plot, suggesting a linear fit. We can attempt a linear fit in *Mathematica* using `LinearModelFit[data,x,x]`, where the first x specifies that a linear term (and implied constant term) should be fitted, and the second x tells *Mathematica* which variable in the preceding expressions should be treated as the independent variable in the fit. The result is shown in Figure 4.8.

If we want to view the R^2 value in *Mathematica*, we can simply use the name we have assigned to our `FittedModel` object (`linearfit` was used in the example code) and use the desired parameter as function input, e.g., `linearfit[“AdjustedRSquared”]`. In this case, our R^2 value is 0.94, not dismal but not spectacular. But the real clue that we’ve missed some structure in our functional fit appears in the residual plot (Figure 4.8). The presence of second-order polynomial structure that is left over after our first-order fit strongly hints that the true order of the polynomial is ~ 3 .

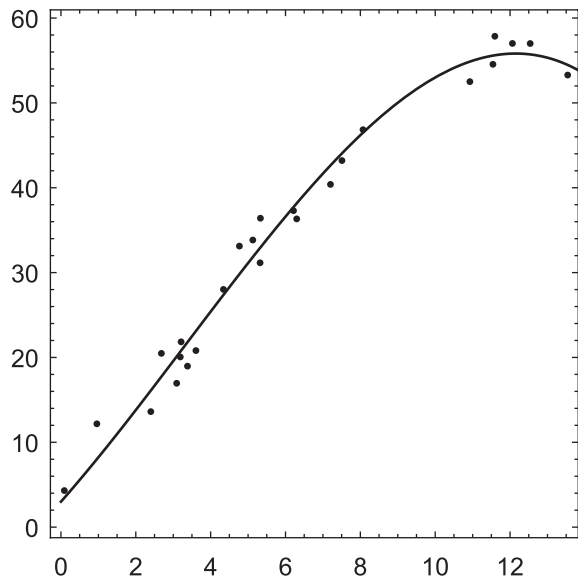


FIGURE 4.7 A pseudorandomly generated data set with normally distributed variations around a third-order polynomial, shown as the solid black line.

* Despite the name, `LinearModelFit` is not restricted to first-order polynomial fitting; the fits are linear in the sense that you provide *Mathematica* with a list of basis functions and it finds the best fit using a single coefficient for each one.

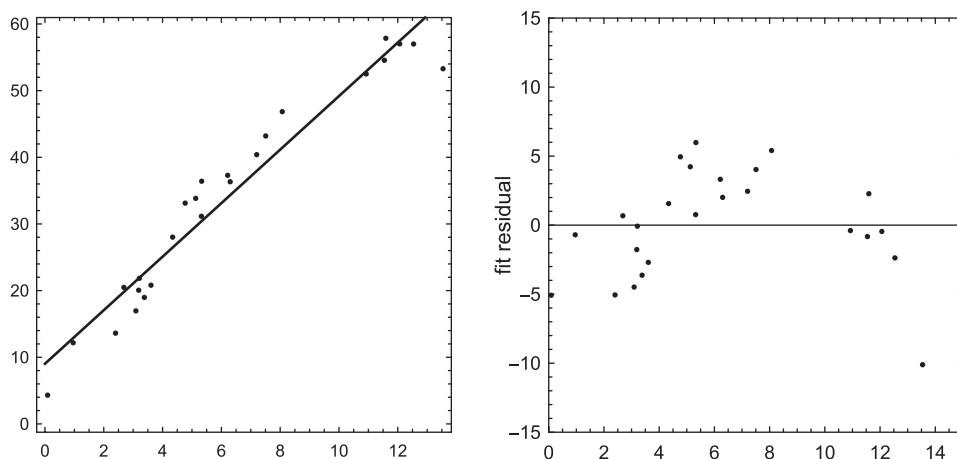


FIGURE 4.8 The first attempt to fit the data set using a line as our model. The left plot shows the line of best fit, while the right plots our residuals, relative to that linear model. Note that the residuals do not appear random, but instead follow a parabolic shape, arcing first upward and then downward.

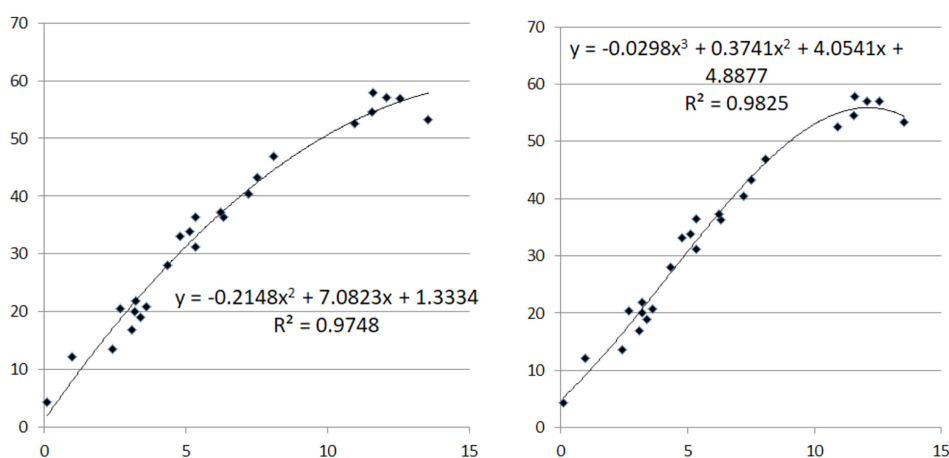


FIGURE 4.9 Excel plots showing second- and third-order fits to the same data set. Note the increasing R^2 values as we increase the order of the polynomial fit.

Using Excel, we can plot our data set and add trendlines using polynomials of orders 2 and 3, as shown in Figure 4.9. While the increasing R^2 is an encouraging sign, it's important to be careful of *overfitting*, or using more parameters than are necessary to describe your data set. After all, if you were given n parameters to fit a data set of n points, you could simply specify every point in your model, with no leftover variance at all – a perfect adjusted R^2 of one! But such a model would be undesirable in at least two ways: it would do nothing to briefly summarize the known data, since your model representation would be just as long as your data set; and it would be unable to predict the outcome of new experiments, unless they used precisely identical test conditions to earlier experiments. Even in the case where you use an overly complex mathematical function to fit your data, you will find that the more parameters you include to reproduce your existing data, the less stable your function will be outside the range of known data, often making it quite useless for extrapolation to new experiments.

How can you avoid overfitting? One method is to monitor your residuals as you continue to add parameters to your fit (Figure 4.10). If you continue to see non-random structure in your residuals, you might be justified in continuing to add fit parameters (or considering a different functional form that will better capture the shape of your underlying data). In physics we sometimes

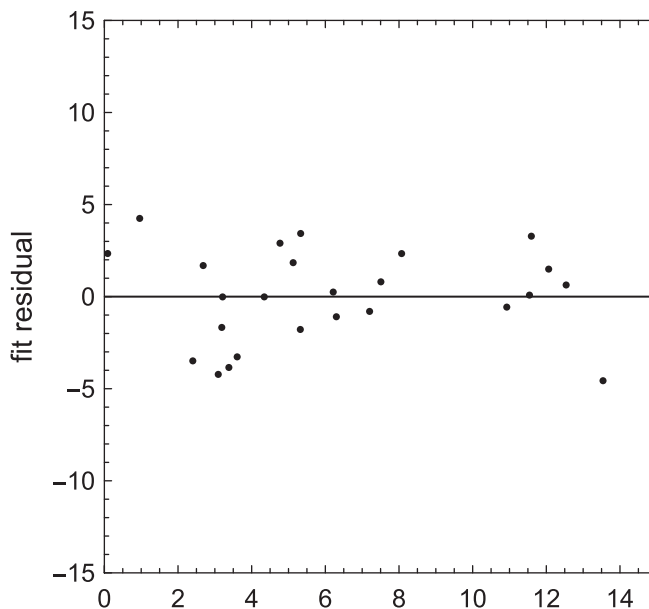


FIGURE 4.10 Residuals from the second-order polynomial fit. Checking residuals for structure is far from foolproof, especially in small data sets. Although the true form of the underlying function is cubic, it is difficult to detect any structure in this set of residuals.

make use of another strategy: using a fit function that is justified by some underlying theory. This has the advantage of guiding you in the choice of fitting function, but if your data are in genuine disagreement with the theory, restricting yourself this way may disguise that disagreement by forcing the data into the theoretical fit.

You can also test each parameter in your expression for overfitting by using a confidence interval test. What should your test hypothesis be? In most cases, a fit parameter equal to zero contributes nothing to the fit (but be careful with fitted exponents!) so you can consider discarding any parameter that lies closer to zero than its own confidence interval. For example, in our example data set above, where the true function was a third-order polynomial, the fit parameters and values for the first five polynomial orders are shown in Table 4.2.

TABLE 4.2

Best-fit parameters for the first five orders of polynomial fit to our test data, whose true functional form was a third-order polynomial. Under each parameter is the value divided by its standard error; values which differ from zero by at least two standard errors are shown in bold

Best-fit coefficients						
(value/standard error)	Constant	x	x^2	x^3	x^4	x^5
Polynomial order						
1	9.02 (5.95)	4.02 (19.2)				
2	1.33 (0.76)	7.08 (12.1)	0.21 (5.39)			
3	4.89 (2.58)	4.05 (3.64)	0.37 (1.91)	0.030 (3.05)		
4	5.01 (2.30)	3.86 (1.94)	0.60 (0.74)	0.067 (0.56)	0.0003 (0.12)	
5	5.99 (2.47)	0.64 (0.16)	2.20 (1.13)	-0.39 (1.04)	0.029 (0.96)	-0.0008 (0.95)

Although nearly every polynomial fit in Table 4.2 has at least one coefficient that is indistinguishable from zero, it isn't until we reach order 4 that the majority of coefficients fail to be significant, a pattern which continues in order 5. While we cannot be definitively sure that the correct fit is the polynomial of order 3, there is a strong hint here that orders 4 and 5 might be overfitting.

4.11 Measurement Errors and χ^2 (also known as chi square)

So far, our fits have treated all of the input data as though they are of equal quality; but in a real experiment, we often want to combine measurements taken under different conditions, or with different instruments, which naturally have different errors of their own. These data should not be given equal weight, but instead more weight should be given to higher-quality data. To obtain the best weighted fit, each point should be weighted inversely by its *variance*, which for a normal distribution is σ^2 . For example, a weighted mean would be given by

$$\mu_{\text{weighted}} = \frac{\sum_{i=1}^n (x_i / \sigma_i^2)}{\sum_{i=1}^n (1 / \sigma_i^2)} \quad (4.18)$$

and the standard error on that weighted mean by

$$\text{SEM}_{\text{weighted}} = \left(\sum_{i=1}^n \frac{1}{\sigma_i^2} \right)^{-0.5} \quad (4.19)$$

To extend this idea to our least-squares function fitting, we need to weight deviations from the fit less for less-reliable data. In un-weighted least-squares, the metric for how discrepant any given point is from the fit is the squared difference between the observed value and the model value; to add inverse variance weighting, we will divide each difference by the squared error on the observation:

$$\chi^2 = \sum_{i=1}^n \frac{(x_i - m_i)^2}{\sigma_i^2} \quad (4.20^*)$$

Minimizing this χ^2 will then have the effect of finding the best-fit parameters for our model. In *Mathematica*, weighted fitting is implemented by adding `Weights → {w1, w2, ..., wn}` to the `LinearModelFit` command.

Interpreting χ^2

If you fitted a model to data and you included your error bars in the fitting process, then the value of χ^2 can tell you something about the quality of your fit and your error bars. Typically, we don't look at χ^2 directly, but at the *reduced* χ^2 , which is χ^2/DOF , where DOF is the number of degrees of freedom from Equation 4.7. Since χ^2 is a sum over all data points, it requires this correction to keep χ^2 from being larger simply from having a larger data set. If no error bars were used in the fitting, then the best-fit model will still be the one with the smallest χ^2 , but without error bars, we cannot predict what that smallest value will be.

On the other hand, if your error bars are a good representation of the actual scatter in your data, then you can expect a reduced $\chi^2 \approx 1$. If your error bars are too small, or your fitting function does a poor job of following your data, or both, then you may see reduced $\chi^2 \gg 1$, since your residuals will appear large relative to your expected residuals. If your error bars are too large, and your fitting function isn't

* In discussions of chi square fitting, you may run across a different definition of the test statistic:

$$\chi^2 = \sum_{i=1}^n \frac{(x_i - m_i)^2}{m_i}$$

This alternative definition is identical to Eq. 4.20 only for the special case of Poisson statistics, where m_i is equal to σ_i^2 , and should only be used to determine the χ^2 for binned data sets (histograms), which naturally follow Poisson statistics.

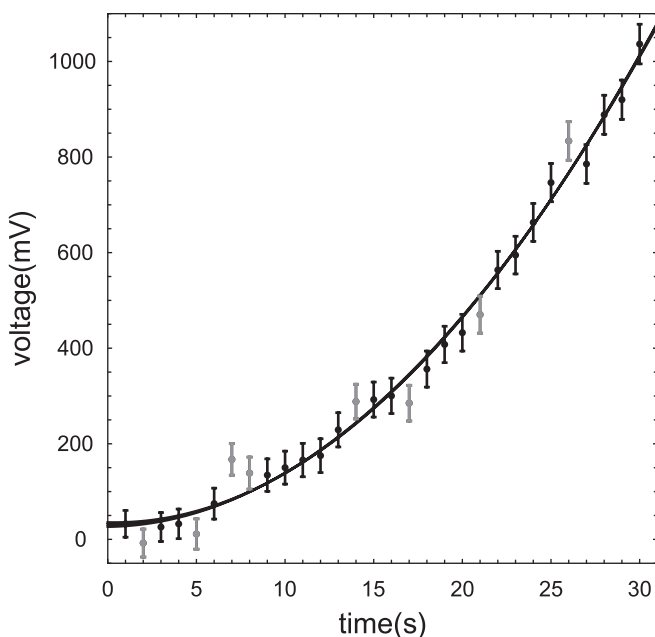


FIGURE 4.11 A simulated data set showing best-fit lines with no weighting, proper weighting, and incorrect weighting (solid black lines). While correctly weighting the fit makes only minor changes to the actual best-fit values in this case, note that the correctly calculated error bars (gray) disagree with the fit for eight of the 30 data points, very close to the expected number of 1σ disagreements (10).

completely incorrect, you may see reduced $\chi^2 < 1$, meaning that your data actually hew more closely to the fitting function than you would have expected, based on your error bars. You can also perform a quick visual check (Figure 4.11) to determine whether your error bars are capturing the true scatter in the underlying data: for normally distributed errors, we expect 33% of our data points to *disagree* with the fitted function by at least one standard error. In a data set of 30 points, there should be only a 1-in-165,000 chance that all of your data will lie within one standard error of the best-fit function, so if you find that your data are all or nearly all in agreement with your fit, it's an excellent bet that you've overestimated the error bars on your individual data points.

Fitting Routines and How to Make Them Work for You

When you need to find a model to fit your data, the most common thing to do is pick out a family of functions (polynomials, exponentials, etc.) and run a fitting algorithm to find the parameters that reproduce your data the best. If everything works according to plan, you'll get physically meaningful parameters (e.g., fitting the slope of a plot of force vs. acceleration to get a cart's inertial mass) with sensible confidence intervals. But just as often, you'll try to fit a model to your data and get parameters that are completely the wrong order of magnitude or get warnings that your fit “failed to converge.” In order to understand these failures, it's worth taking a quick look at how model fitting is done.

Any model fitting routine will require a “goodness-of-fit” estimator to tell it whether it has found the best parameters for your data. The most traditional estimator is actually χ^2 from the earlier section on fit quality and error bar sizes. It has the desirable properties that for simple linear models it has a global minimum at the best-fit parameters and is everywhere continuous and differentiable, which makes it easy for automated routines to find its minimum – they can evaluate χ^2 and its gradient at a point in parameter space, then follow the downward gradient in a series of steps along the χ^2 surface to its lowest point (the method of steepest descent). But there's a possible failure in the plan: what we want is the best fit from all possible parameters, but the method of steepest descent is particularly susceptible to getting stuck in a merely local minimum, a set of parameters that is a better fit than all of its close-by neighbors in parameter space, but a worse fit than some very distant set of parameters.

A related technique that shares only some of the disadvantages of steepest descent is called downhill simplex. This method works better for functions whose derivatives might be unknown or difficult to calculate, so rather than use the calculated gradient at a location to determine which way the algorithm should take a step, we calculate the value of the goodness-of-fit parameter at a variety of small steps in every possible parameter direction (these points form a simplex in parameter space). We then take a step in the direction that reduces χ^2 by the largest amount before repeating the process, flowing downhill along the parameter surface like an amoeba. Unfortunately, this method has the same sort of problems with getting trapped in local minima that the steepest descent method has, although it can more easily adapt its step size and search radius in an attempt to escape from a local minimum and search for something even lower.

Regardless of your search method, any fitting algorithm can fail if the χ^2 surface is complex enough. Although a search for a best-fit line will always converge, as we move to more complicated parameterizations (exponential, sinusoidal, or even just polynomial), we have an increased chance that our search algorithm will miss the “correct” answer and just keep taking steps off into an unrealistic region of parameter space.

To have the best chance of finding good parameter fits:

- *Choose an appropriate fitting function.* You should plot your data before proceeding with a fit, and if the shape of the data suggests a particular functional form (linear, exponential, sinusoidal, etc.) then choose something appropriate, or use a fitting function based on theory. Even the best fitted parameters will provide a poor fit and worse predictions if they are calculated with a function that does not capture the behavior of your data.
- *Start with a good estimate.* Most iterative fitting programs will allow you to set the starting values for each search parameter. (In *Mathematica*, this can be done by placing each parameter in a list with its starting value, e.g., {a, 500} would start the search for parameter *a* at value 500.) If there is any way for you to estimate a realistic value for your parameter, you should include it as a starting point. This will dramatically decrease the odds that your fitting program will get caught in a local minimum or run off to infinity.
- *Lock down any parameters you are not using.* If you’re using a standard expression for your fitting function, but you don’t need certain parts of it, set those to be fixed values rather than allowing your program to fit unnecessary parameters. You might manually replace those parameters with numbers, or some fitting programs (such as Origin) allow you to specify certain parameters to have a fixed value rather than being fitted. This will prevent possible fitting errors from changes in those unwanted parameters.
- *Don’t be afraid to experiment with alternative fits.* If you see something suggestive in the shape of your data or your residuals after a fit, try changing your fit function or the starting points of your parameters. Even if you have a strong theoretical basis for picking, say, a linear function for your fit, the quadratic outliers or sinusoidal pattern might be telling you something about systematic errors or unmodeled details of your real physical system.

Outliers and Outlier Rejection

Since we expect a certain level of disagreement between our data and our fitted model, how can we define an *outlier*, a point which, due to an error not present in the other data, lies anomalously far from the correct value? If there are truly outliers in the data, we would not wish to include them in the fit – since they are, in essence, drawn from a different underlying distribution of values, we cannot include them in our fit at the same weight as the other points without changing our fit in severe and often undesirable ways.

Sometimes, a simple statistical test can detect outliers and allow you to exclude them; for example, when combining multiple astronomical images taken of the same area of the sky, the pixels corresponding to one spot on the sky are often subjected to a “three-sigma clip,” which removes any pixel whose value is more than three standard deviations away from the mean of the pixels at the same position in the other images. For normally distributed errors, we would expect this to be true of only 0.3% of non-outlier pixels, so for a stack of 20 images, there is a 94% chance that all 20 pixels at a given spot will meet this criterion, while it’s almost certain to correctly reject pixels that have been hit by cosmic rays (giving a

very high value) or pixels with unusually low response due to charge traps or other unrecognized issues with the camera. But bear in mind that if we have 1000 images to combine, then this criterion would be expected to eliminate three legitimate pixels out of that 1000. So there is a cost to using this clipping procedure to eliminate bad pixels, since it eliminates good pixels as well, but we might consider it worthwhile because the rate of bad pixels is relatively high and the inclusion of those bad pixels would impose a significant cost in the final image (by shifting the calculated mean to a distinctly incorrect value).

In another experiment, this type of clipping might be less appropriate: suppose that you were running a particle detection experiment where you were trying to measure how likely one rare decay was, compared with another more common decay. If you applied a sigma-based rejection criterion to each data point, you would find yourself rejecting more legitimate rare decays than legitimate common decays, simply because they would lie farther from the mean of the typical random sample.

These statistical methods are effective for identifying the occasional outlier whose cause is unknown. When running an experiment, you may consider treating data differently, or even discarding them, when there is a clear, articulable reason that they do not agree with your other measurements, for example:

- You discover that a piece of equipment was malfunctioning.
- The apparatus was warming up, cooling down, or otherwise unstable.
- Your records of that batch are flawed or incomplete.
- You opened a new box of supplies and find that all of your measurements have shifted.
- One series of data disagrees with an otherwise strong empirical trend.

In these cases, you must decide for yourself whether to eliminate the data from your report entirely, report them but exclude them from your analysis, include them in your analysis but at a lower weight, or just allow them to affect your results as they may. Excluding data points because they do not fit your expectations or your trend would be considered unethical, but real-world experimentation is full of tweaks to experimental setups, iterative improvements, bug fixes, and other realistic reasons to discard some or all of your data.

How can you make sure that you're taking a responsible approach to data collection, rather than cherry-picking your data to achieve your desired result?

- *Set data collection goals ahead of time and stick with them:* it's tempting to stop your data collection when you see the trend you were predicting, or keep going because you haven't seen anything yet, but this is one of the easiest ways to accidentally bias your data in favor of your desired outcome.
- *Conduct a blind analysis whenever possible:* in order to avoid confirmation bias (the natural tendency to weight new evidence by how well it confirms your preconceptions), many large collaborations use blind analyses, where a random offset is added to the data (and securely recorded elsewhere) so that the analysis team won't be able to tell whether their answer agrees with, say, the known speed of light, or the expected value of Hubble's constant or the gyromagnetic ratio of the electron. Since those numbers are so well known, it would be very tempting to keep seeking out new errors or conducting more complicated analyses until your answer matches the "right" answer, and then stop your work. This would make it nearly impossible for you to actually contribute any independent knowledge about the value you are purportedly measuring! Blinding allows an analysis team to truly put in its best effort without any bias toward the library value.
- *A new discovery, or at least a new kind of failure?* Consider that your "unusual" data points might be a genuine signature of a new unrecognized physical effect – possibly the most important clues that an experimenter can find! – or that they might point you to flaws in your experimental design or analysis that you can correct.

4.12 What to Do When Something Goes Wrong

Despite your best efforts, sometimes your experiment will not proceed according to plan – you'll get a result that you know to be incorrect, not merely because it disagrees with your expectations, but because

you recognize, either during or after the experiment, that you did not carry the experiment out correctly. You might have used the equipment incorrectly, left out a key measurement that makes it impossible to analyze your results, or lost all record of your data points.

When this happens, the way out is simple: you must not report data that you know to be incorrect without commenting on them. If you can, replace the data by repeating the experiment correctly from the point where it went wrong, in consultation with your lab instructor or research supervisor. Sometimes this is a simple matter of measuring a forgotten parameter, or carefully setting up an experiment so that you can take a few more measurements; take care with your new data, since you might introduce systematic errors by carrying out the experiment at a different time, with different equipment, or simply by using the apparatus more than once.

If recovering the experiment is not possible, document the failure as clearly as you can and explain how it affects your results: which parts of the experiment can still be completed and which cannot, how you would have proceeded if the experiment had been successful, and what you would need to do differently to correct this issue in the future. A careful consideration of the failure points of an experiment can be a big help to the next person who attempts that measurement.

4.13 Homework Problems

- For each experiment and effect, figure out whether it would contribute random error, systematic error, or both types:
 - Near a vibration-sensitive laser interferometry experiment: daily traffic rises and falls
 - Near a DC electronic experiment: 60 Hz electrical interference from nearby wires
 - Above a shielded cosmic ray detector: seasonal rise and fall in water table
 - During a sensitive light-measuring experiment: stray light leaking in from the room
 - During a photon-counting experiment: changing the gain setting so that one photon appears as ten counts (instead of one)
- Why can't repeated trials reduce systematic error? How *can* systematic error be reduced?
- In an experiment dominated by random error, you take automated readings from a highly accurate voltmeter and average them in groups of 20. If the settings are changed to average readings in groups of 80, how does this affect the mean, the standard deviation (σ), and the standard error on the mean (SEM), compared with the earlier measurements?
- An experiment that you read about quoted a mean value for Hubble's constant of $71 \text{ km/s} \pm 3$ (systematic) ± 4 (random). If you expanded the sample from that experiment by a factor of 25, how would you expect each of those error estimates to change? What would be the old and new total errors, if you assume that the systematic and random errors are uncorrelated?
- What would be the standard way of reporting the following raw values and uncertainties, using the correct number of significant figures?
 - 75.8 ± 17.5
 - 110.7 ± 43
 - 20 ± 1.73
 - 3554 ± 75
- For the following measured and theoretical values, determine whether they agree or disagree at the 95% level, given the attached uncertainties.
 - $10 \pm 3 \text{ V}$ (1σ , measured) and $11 \pm 1.2 \text{ V}$ (1σ , theoretical)
 - $1.6 \pm 0.5 \times 10^8$ (95%, measured) and $9 \pm 5 \times 10^7$ (95%, theoretical)
 - $150 \pm 20 \text{ m/s}$ (95%, measured) and $140 \pm 20 \text{ m/s}$ (95%, theoretical)
 - $4900 \pm 300 \Omega$ (1σ , measured) and $4550 \pm 70 \Omega$ (1σ , measured)
- For many particle detection experiments, the signals are recorded using *photomultiplier tubes*, which can detect a single incoming photon by creating a cascade of electrons to amplify the

signal. Since the experiment counts single photons arriving during some time interval, the data should obey the *Poisson distribution*:

$$P(k) = \frac{\mu^k e^{-\mu}}{k!}$$

where μ is the mean number of events per interval (say, photons per second) and $P(k)$ is the probability that exactly k events will be detected in any particular time interval.

- (a) If the mean number of photons detected per second is 5, calculate the detection probabilities for $k = 1, 2, 3, \dots, 10$. (The Excel EXP() and FACT() functions should be useful for this.)
 - (b) Repeat the previous calculation for $\mu = 25$ and $\mu = 100$, calculating k for each integer from 1 to 2μ . (Excel will suffer simultaneous under- and over-flow for $k > 150$, but you should be able to discern the pattern by then.)
 - (c) How does the plot of the Poisson distribution for $\mu = 100$ compare with a plot of the normal distribution for $\mu = 100$, $\sigma = 10$? When using Excel to calculate the values of the normal distribution, you may need to add extra parentheses to ensure that the argument of your EXP() function is always negative.
8. Suppose you are working on a radioactive decay experiment, measuring decays per minute over the course of ten minutes (ten data points). If your fitting function is

$$N(t) = N_0 e^{-at}$$

how many degrees of freedom does your fit have? How wide (measured in standard errors) should your 95% confidence interval be?

9. During an experiment to check Coulomb's Law $\left(F = \frac{kq_1q_2}{r^2} \right)$, you measure the following values:

$$q_1 = +5 \pm 0.7 \text{ mC} \quad q_2 = -7 \pm 1.1 \text{ mC} \quad r = 0.05 \pm 0.01 \text{ m}$$

You can assume the error on $k = 9 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2}$ is much smaller than the other errors in the

experiment. If your force gauge reads $0.63 \mu\text{N}$, does that measurement support Coulomb's Law or contradict it?

10. For $y = Ax^n z^m$, derive equation 4.14.

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FIGURE 4.12 Dr. Paul Thorman.